

Modeling: Introduction

DATA 202 21FA

Logistics

- Midterm 1 Online Portion is on Moodle
 - Don't forgot to copy-paste responses and completion code to Moodle
 - Closes Wed evening
- Project Milestone 2 due Friday
- No quiz or Discussion this week

Q&A!

What kinds of things can functions return?

Anything you can assign to a variable. (numbers, strings, data frames, plots, ...)

Are `as.numeric` and `as.integer` the same?

```
as.numeric("1.7")
```

```
[1] 1.7
```

```
as.integer("1.7")
```

```
[1] 1
```

```
as.integer("One Point Seven")
```

```
Warning: NAs introduced by coercion
```

```
[1] NA
```

2016 Honda Odyssey

EX-L 4dr Minivan (3.5L 6cyl 6A)



Mileage	37,183
Condition	Outstanding
Exterior	Modern Steel Metallic

Your appraisal As of 10/26/2019

	Trade-In	Private Party	Dealer Retail
Email report	\$21,645	\$23,731	\$25,666
National Base Price ⓘ	\$19,676	\$21,694	\$23,484
Color Adjustment ⓘ	-\$43	-\$48	-\$52
Regional Adjustment ⓘ	\$103	\$114	\$123
Mileage Adjustment ⓘ	\$680	\$680	\$680
Condition Adjustment ⓘ	\$1,229	\$1,291	\$1,431
Value	\$21,645	\$23,731	\$25,666

What data frame does Edmunds have? Sketch an example. (What are the columns? What does each row represent?)

Missing Data

- Tools so far very good when we have all the data we want.
- Usually we *don't*-- or even we *can't*.

Different kinds of missingness:

1. How much will this car / house sell for?
2. What will Bitcoin trade for at end-of-day today?
3. What will this addition do to the price of my house?
4. I feel rotten. Is it Covid?
5. What word will the user type next on their keyboard?
6. Will this home-buyer default on their loan?

Discussion:

1. For each of these, discuss **what data is missing** (*why?*).
2. Which of these situations are similar to each other?

Some terminology

- **Supervised learning**: for each item independently, fill in an unobserved variable
 - **Regression**: fill in a *number*
 - **Classification**: fill in a *choice*
- **Unsupervised learning**: identify *relationships* between items
- **Forecasting**: predict how a sequence will continue (future observations)
- **Statistical inference**: fill in summary statistics (we wanted a population but only got a sample)
- **Causal inference**: fill in counterfactuals (what if?)

Predictive Analytics

Mostly *supervised learning* (*regression* and *classification*), some *forecasting*.

- A powerful tool to turn data into action.
- It works because God made the universe predictable (and successful prediction rewarding)
- **Need for wisdom:** It can be used for great good and great harm

Power of Predictive Modeling

- **Medicine**: wearable monitor for seizures or falls, detect malaria from blood smears, find effective drug regimens from medical records
- **Drug Discovery**: predict the efficacy of a synthesis plan for a drug
- **Precision Agriculture**: predict effect of micro-climate on plant growth
- **Urban Planning**: forecast resource needs, extreme weather risks, ...
- **Government**: classify feedback from constituents
- **Retail**: predict items in a grocery order
- **Recommendation systems**: Amazon, Netflix, YouTube, ...
- **User interfaces**: gesture typing, autocomplete / autocorrect

and so much more...

The universe is surprisingly predictable

- God created the world with actionable structure
 - We gradually learn how to perceive that structure and act within it.
 - The better our perceptions align with how the universe is structured, the better our actions
 - We can discover that structure by learning to be less surprised by what we see (= predicting our perceptions)
- Perceptions are thus both accurate and fallable.

Predictive modeling technology: Need for wisdom

- Potential for great good
- But also great harm:
 - Lack of **fairness** in facial recognition, sentencing, lending, job applicant scoring, ...
 - Lack of **transparency** in how "Big Data" systems make conclusions
 - Lack of **privacy** as data is increasingly collected and aggregated
 - Amplification of extreme positions in social media, YouTube, etc.
 - Oversimplification of human experience
 - Hidden human labor
 - Illusion of objectivity
 - ...!

Wednesday

Q&A

| How will we code predictive models?

- We'll use `tidymodels`, a toolkit like `tidyverse`. Preview today, practice Friday.
- We'll need all our `tidyverse` skills to understand our data before modeling it and to visualize our models.

| What are the limits? Can we predict everything?

- Silicon Valley: "Yes!"
- Wisdom: "No!"
 - What failure to predict life outcomes can teach us
 - Predictions can be inaccurate and biased, with disparate impacts on vulnerable people.

Different kinds of missing data

- Missing *items*: entire rows not in your data.
 - Some people didn't fill out your survey.
 - You only know about people who *did* visit your website, not those that didn't. etc.
- Missing *observations*: row present, but some observations missing
 - someone added a product to their cart but didn't buy it, so you don't know their address.
 - only some people got an expensive diagnostic test. etc.

Why not just ignore what's missing?

- Ignoring missing data leads to **selection bias** and related biases (see catalogofbias.org).
- Missing *observations* can be exactly the info you need for making a decision, e.g., what price to list your product for.
- Implications:
 - **Never** `drop_na()` without explanation.
 - "The map is not the territory" -- data $\neq$ objective reality

Examples of Supervised Learning

Regression Example: Home Sales

From Ames, Iowa home sales, 2006-2010. (De Cock, 2011)

Lot_Area	Bldg_Type	Gr_Liv_Area	Garage_Cars	Sale_Price
12546	OneFam	1440	2	182.9
2645	Twnhs	1586	2	170.0
15312	OneFam	1138	2	148.0
8544	Duplex	1040	2	81.4
12677	TwnhsE	1518	2	274.0

(2412 total rows)

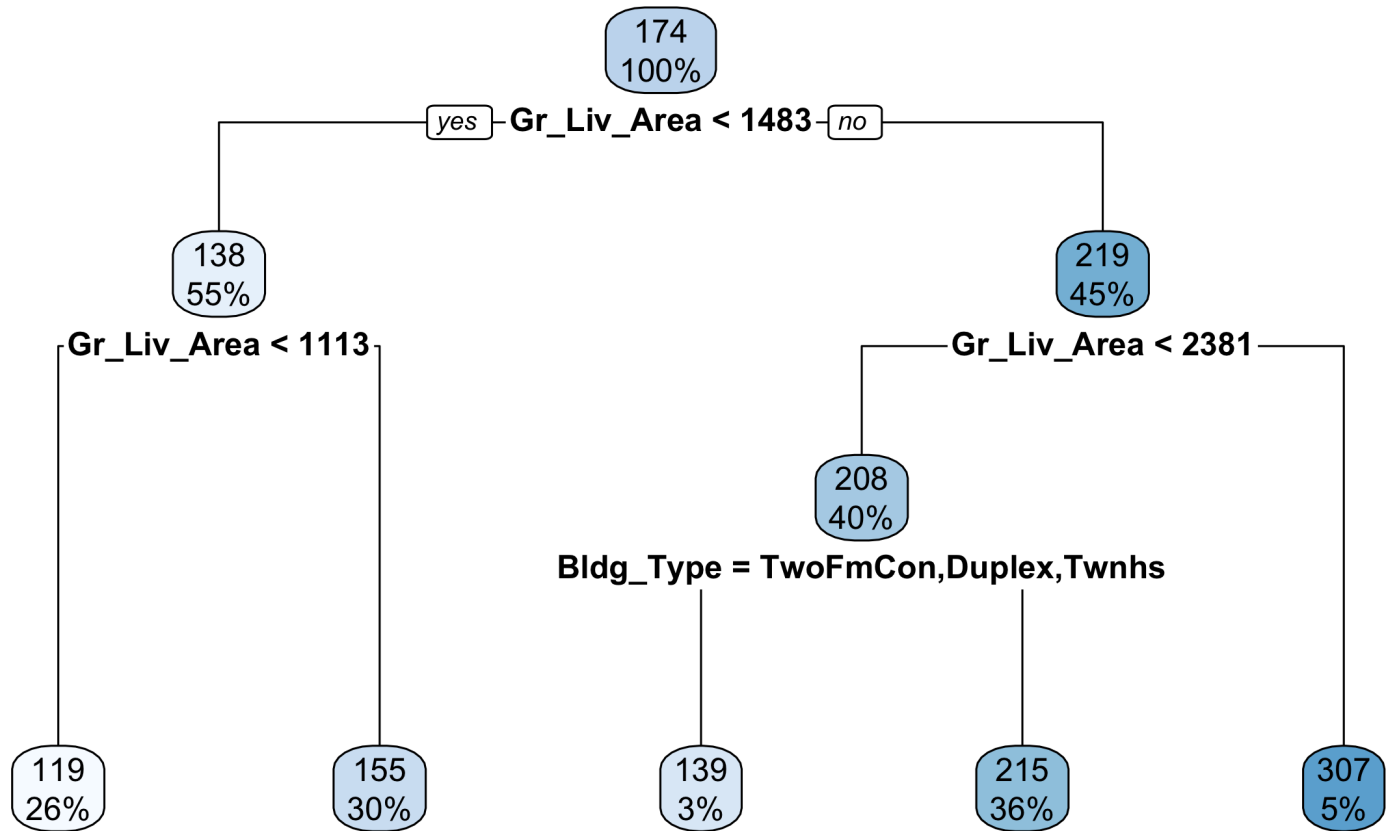
- *y*: response variable (aka *outcome*, *dependent variable*):
Sale_Price
- *X*: features (aka *predictors*, *covariates*, etc.): everything else

Note: *X* is much easier to measure than *y*

Model

Predictions

Model code



Were those predictions good?

```
metrics <- yardstick::metric_set(rsq_trad, mae, mape, rmse)
decision_tree_fit %>%
  predict(ames_test) %>%
  bind_cols(ames_test) %>%
  metrics(truth = Sale_Price, estimate = .pred) %>%
  select(-.estimator) %>% knitr::kable()
```

.metric	.estimate
rsq_trad	0.5254953
mae	35.3756288
mape	21.1170598
rmse	50.5840333

- Traditional R^2 (fraction of variance explained)
- MAE: Mean Absolute Error ("predictions are usually off by \$xxx")
- MAPE: Mean Absolute Percent Error ("predictions are usually off by yy%")

Classification example: Can a blood test diagnose autism?

We'll use an example from a 2017 PLOS Computational Biology paper

```
autism
```

```
# A tibble: 206 × 26
  Group Methion. SAM SAH `SAM/SAH` `% DNA methylatio... `8-OHG`
  <chr>      <dbl> <dbl> <dbl>      <dbl>      <dbl>      <dbl>
1 ASD        17.3  56.2  15.2        3.69        3.35      0.055
2 ASD        14.9  37.2   7.58        4.91        3.04      0.045
3 ASD        15.9  37.9   9.87        3.84        2.81      0.058
4 ASD        18.7  79.2  24.5        3.23        4.24      0.085
5 ASD        21.5  77.6  19.2        4.04        3.49      0.041
6 ASD        18.1  67.6  12.8        5.30        3.01      0.156
# ... with 200 more rows, and 19 more variables: Adenosine <dbl>,
# Homocysteine <dbl>, Cysteine <dbl>, Glu.-Cys. <dbl>,
# Cys.-Gly. <dbl>, tGSH <dbl>, fGSH <dbl>, GSSG <dbl>,
# fGSH/GSSG <dbl>, tGSH/GSSG <dbl>, Chlorotyrosine <dbl>,
# Nitrotyrosine <dbl>, Tyrosine <dbl>, Tryptophane <dbl>.
```

We have 3 kinds of data about 206 children:

1. The outcome (**Group**): ASD (diagnosed with ASD), SIB (sibling not diagnosed with ASD), and NEU (age-matched neurotypical children, for control)

```
autism %>% group_by(Group) %>% summarize(n = n()) %>% knitr::kable
```

Group	n
ASD	83
NEU	76
SIB	47

1. The outcome (Group): ASD, SIB, NEU
2. Concentrations of various metabolites in a blood sample:

```
autism %>% select(-1, -last_col())
```

```
# A tibble: 206 × 24
```

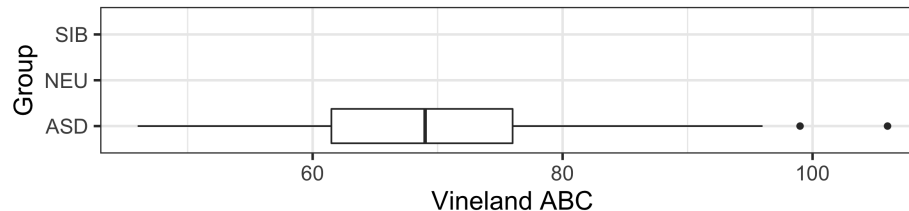
	Methion.	SAM	SAH	`SAM/SAH`	`% DNA methylation`	`8-OHG`
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	17.3	56.2	15.2	3.69	3.35	0.055
2	14.9	37.2	7.58	4.91	3.04	0.045
3	15.9	37.9	9.87	3.84	2.81	0.058
4	18.7	79.2	24.5	3.23	4.24	0.085
5	21.5	77.6	19.2	4.04	3.49	0.041
6	18.1	67.6	12.8	5.30	3.01	0.156

```
# ... with 200 more rows, and 18 more variables: Adenosine <dbl>,  
# Homocysteine <dbl>, Cysteine <dbl>, Glu.-Cys. <dbl>,  
# Cys.-Gly. <dbl>, tGSH <dbl>, fGSH <dbl>, GSSG <dbl>,  
# fGSH/GSSG <dbl>, tGSH/GSSG <dbl>, Chlorotyrosine <dbl>,  
# Nitrotyrosine <dbl>, Tyrosine <dbl>, Tryptophane <dbl>,  
# fCystine <dbl>, fCysteine <dbl>, fCystine/fCysteine <dbl>,  
# % oxidized <dbl>
```

1. The outcome (**Group**): ASD, SIB, NEU
2. Concentrations of various metabolites in a blood sample
3. For the ASD children only, a measure of life skills ("Vineland ABC")

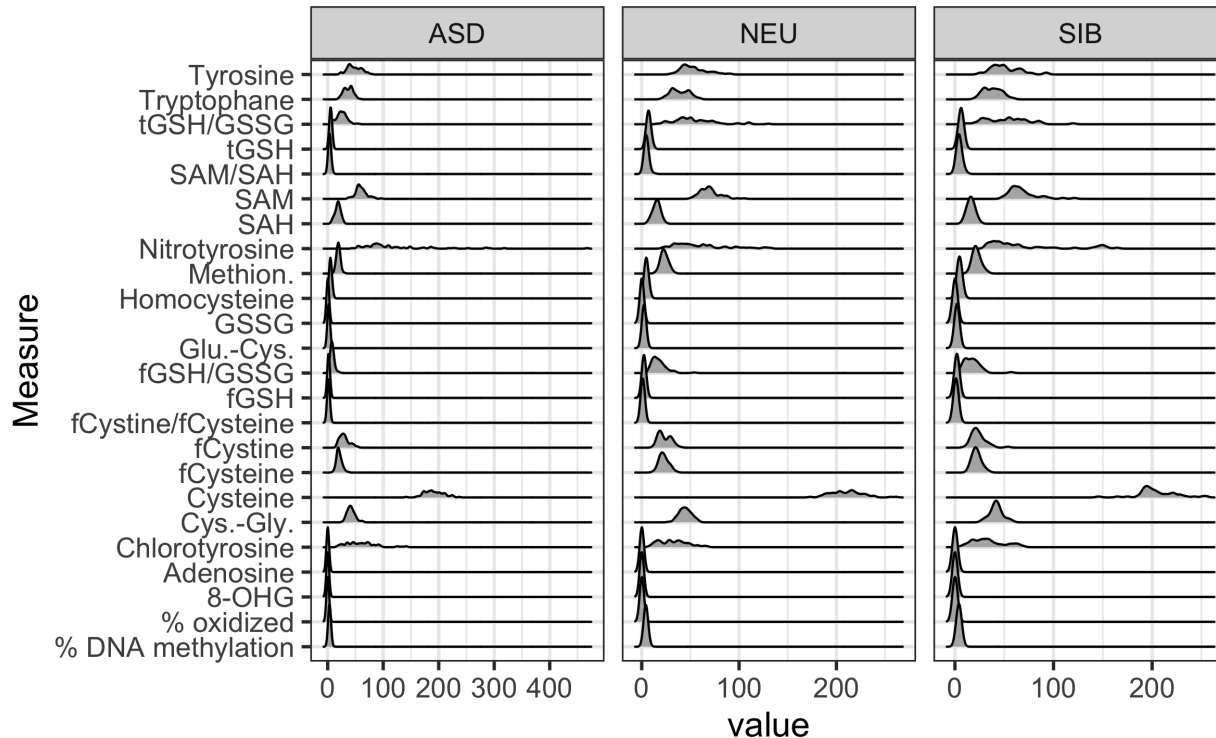
```
autism %>%  
  ggplot(aes(x = `Vineland ABC`, y = Group)) + geom_boxplot()
```

Warning: Removed 159 rows containing non-finite values (stat_boxplot).



Exploratory Data Analysis (EDA)

What do these metabolites look like?

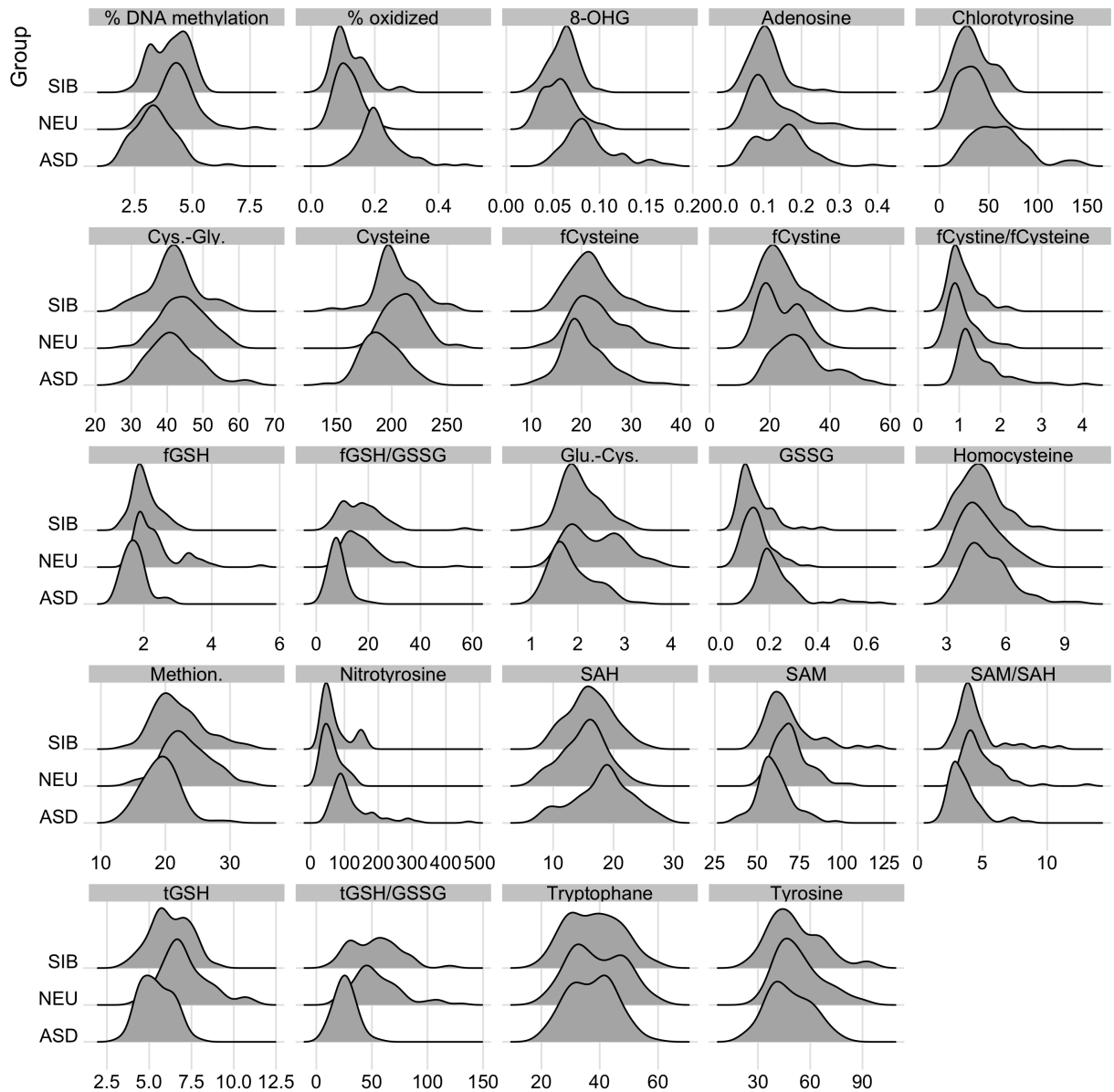


code for the previous plot:

```
library(ggribes)
autism %>%
  select(-`Vineland ABC`) %>%
  pivot_longer(-Group, names_to = "Measure") %>%
  ggplot(aes(x = value, y = Measure)) +
  geom_density_ridges() +
  facet_wrap(vars(Group), scales = "free_x")
```


EDA

Better question: **Can these metabolites help us distinguish autism?**



code for previous plot:

```
autism %>%  
  select(-`Vineland ABC`) %>%  
  pivot_longer(-Group, names_to = "Measure") %>%  
  ggplot(aes(x = value, y = Group)) +  
  geom_density_ridges() +  
  facet_wrap(vars(Measure), scales = "free_x") +  
  theme_ridges()
```

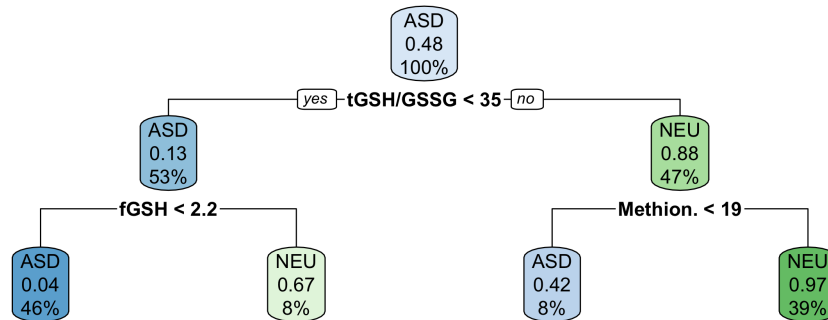
Can we predict ASD vs non-ASD from metabolites?

- Let's start by (1) ignoring the behavior scores (that's an *outcome*) and comparing just ASD and NEU.
- We need to drop SIB and encode `Group` as a factor.

```
data <- autism %>%  
  select(-`Vineland ABC`) %>%  
  filter(Group != "SIB") %>%  
  mutate(Group = as_factor(Group))
```

Decision Tree *Classification*

```
spec <- workflow() %>% add_recipe(  
  recipe(Group ~ ., data = data)) %>%  
  add_model(decision_tree(mode = "classification")) %>% set_engine(  
model <- spec %>% fit(data)
```



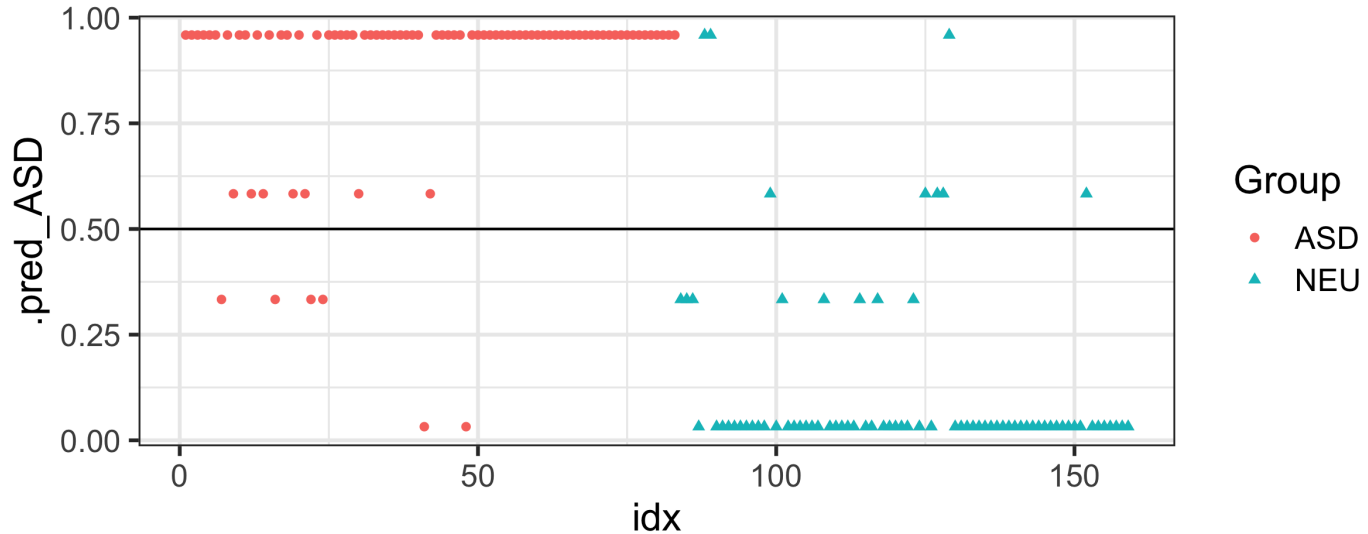
What do the *predictions* look like?

```
model %>% predict(data, type = "prob")
```

```
# A tibble: 159 × 2  
  .pred_ASD .pred_NEU  
    <dbl>    <dbl>  
1     0.959     0.0411  
2     0.959     0.0411  
3     0.959     0.0411  
4     0.959     0.0411  
5     0.959     0.0411  
6     0.959     0.0411  
# ... with 153 more rows
```

Were those predictions *good*?

```
model %>%  
  predict(data, type = "prob") %>%  
  bind_cols(data) %>%  
  mutate(idx = row_number()) %>%  
  ggplot(aes(x = idx, y = .pred_ASD, color = Group, shape = Group)) +  
    geom_hline(yintercept = .5) +  
    geom_point()
```



Quantifying that:

```
metrics <- yardstick::metric_set(accuracy, sensitivity, specificity)
model %>%
  predict(data, type = "class") %>%
  bind_cols(data) %>%
  metrics(truth = Group, estimate = .pred_class)
```

```
# A tibble: 3 × 3
  .metric .estimator .estimate
  <chr>    <chr>         <dbl>
1 accuracy binary         0.912
2 sens     binary         0.928
3 spec     binary         0.895
```


Classification Metrics

	Event happened	No event happened
Event predicted	True positive	False positive (Type 1 error)
No event predicted	False negative (Type 2 error)	True negative

Classification Metrics

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No event predicted	False negative (Type 2 error)	True negative

- **Accuracy** (% correct) = $(TP + TN) / (\# \text{ episodes})$
- **False negative** ("miss") **rate** = $FN / (\# \text{ actual events})$
- **False positive** ("false alarm") **rate** = $FP / (\# \text{ true non-events})$

Classification Metrics

	Event happened	No event happened
Event predicted	True positive	False positive (Type 1 error)
No event predicted	False negative (Type 2 error)	True negative

- **Accuracy** (% correct) = $(TP + TN) / (\# \text{ episodes})$
- **False negative** ("miss") **rate** = $FN / (\# \text{ actual events})$
- **False positive** ("false alarm") **rate** = $FP / (\# \text{ true non-events})$
- **Sensitivity** ("true positive rate") = $TP / (\# \text{ actual events})$
 - Sensitivity = $1 - \text{False negative rate}$
- **Specificity** ("true negative rate") = $TN / (\# \text{ actual events})$
 - Specificity = $1 - \text{False positive rate}$
- [Wikipedia article](#)